

Muhua (黄牧华) Huang

muhua@stanford.edu · [muhua-h.github.io](https://github.com/muhua-h) · [LinkedIn](#) · [Google Scholar](#)

Computational social scientist and Stanford Organizational Behavior PhD researcher studying how AI systems represent human traits and reshape collective behavior. I build large-scale LLM-agent experiments and psychometric methods to investigate human-AI collaboration, culture, values, and emergent social dynamics.

EDUCATION

Stanford Graduate School of Business, PhD in Organizational Behavior	2025 – 2030
University of Chicago, MA in Computational Social Science, GPA: 3.9/4.0	2023 – 2025
University of British Columbia, BA in Computer Science & Psychology (Honours), GPA: 4.2/4.3	2018 – 2023

RESEARCH & INDUSTRY EXPERIENCE

PhD Student Researcher | Stanford Graduate School of Business 2025 – Present

- Built a large-scale multi-agent coordination experiment with approximately 375,000 GPT-4o trials across networks of 2–50 agents, replicating a human category-formation paradigm.
- Identified systematic human-AI differences in category boundaries and showed that constraining agents to human labels reduced performance by 12.7%.

Student Researcher | Google DeepMind Jun – Sep 2026

Research Intern | Microsoft Research Asia Jul – Oct 2024

Master's Researcher | University of Chicago 2023 – 2025

- Developed a psychometrically validated framework for assigning Big Five personalities to LLM agents across GPT-3.5, GPT-4, and Claude 3.
- Collected data from 350 participants and showed that assigned personalities predict agents' moral reasoning, risk-taking, and cooperation; published in *Personality Science*.

Research Assistant | University of British Columbia 2020 – 2023

- Applied SEM, ML/NLP, simulation, and eye-tracking across quantitative psychology, social psychology, and human-centered AI projects, contributing to three peer-reviewed publications.

SELECTED PUBLICATIONS

- Huang, M.**, Zhang, X., Soto, C., & Evans, J. (2026). Designing AI-Agents with Personalities: A Psychometric Approach. *Personality Science*.
- Huang, M.**, & Evans, J. (2026). Institutions as cached computation for resource-rational negotiation. *Behavioral and Brain Sciences*, 49, e144.
- Bai, Y., Duan, S., **Huang, M.**, et al. (2026). IROTE: Human-like Traits Elicitation of Large Language Model via In-Context Self-Reflective Optimization. *AAAI*.
- Savalei, V., & **Huang, M.** (2025). Fit Indices Are Insensitive to Multiple Minor Violations of Perfect Simple Structure in Confirmatory Factor Analysis. *Psychological Methods*.

METHODS & RECOGNITION

Methods: LLM agents, generative agent-based modelling, psychometrics, structural equation modelling, social network analysis, ML/NLP, behavioral experiments, RAG, Python, R, SQL.

Selected recognition: Stanford GSB Doctoral Fellowship; Stanford EDGE Doctoral Fellowship; UChicago Outstanding Thesis Award; Microsoft Research Asia Outstanding Intern Award; OpenAI Researcher Access Program.